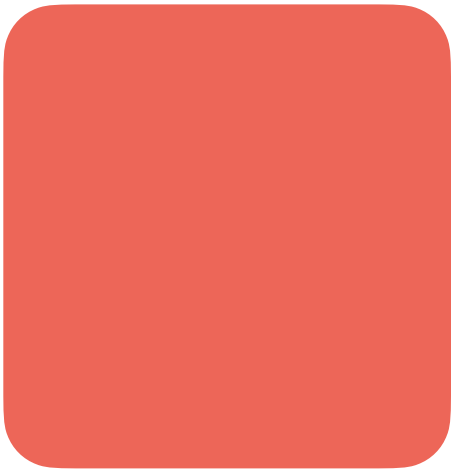
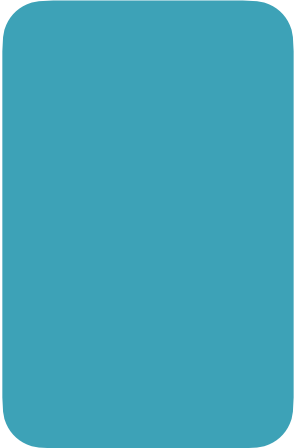


Wasserstein Gradient









**First variation of functional —
“potential”/ driving force field**

mass flux—how much mass moves through each point

Gradient of that potential — local velocity
vector, points to direction of max change of $\mathcal{F}$

Time derivative of the density — how
the distribution changes over time



Converts flux into net inflow

Back to Continuous Time

- One can show that as $\tau \rightarrow 0$, the solution of

$$\rho^{k+1} \in \arg \min_{\rho} \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho^k)$$

- Corresponds to the solution of:

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right)$$

Time derivative of the density — how the distribution changes over time

Converts flux into net inflow/outflow

First variation of functional — “potential”/ driving force field

Gradient of that potential — local velocity vector, points to direction of max change of $\mathcal{F}$

mass flux — how much mass moves through each point

- Define the **Wasserstein Gradient** $\nabla_{\mathbf{W}} \triangleq \nabla \cdot \left(\rho \nabla \frac{\partial(\cdot)}{\partial \rho} \right)$, then gradient flow can be written as: $\partial_t \rho_t = \nabla_{\mathbf{W}} \mathcal{F}(\rho_t)$

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